

Name:
Z number:

Total Points: 26

COP 3410-001: Data Strc Algrthm Anlys Python

FLORIDA ATLANTIC UNIVERSITY

Binary Trees and Traversals

Worksheet

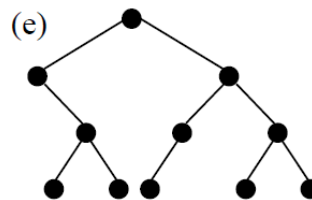
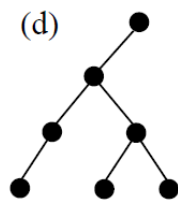
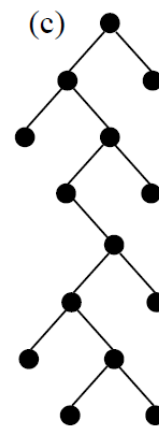
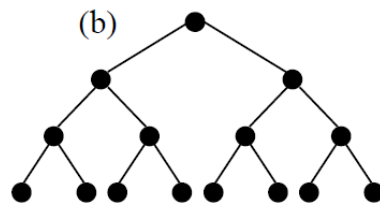
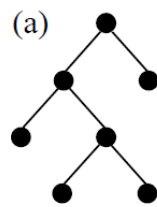
NOTE:

- Each question carries 1 point.
- Submit the completed worksheet in the form of a PDF.
- The file name must be *YourName_ Binary Trees and Traversals Worksheet.pdf*

1- Given a binary tree of size 76, what is the minimum number of levels it can contain? What is the maximum number of levels? (1 points)

2- What is the maximum number of nodes possible in a binary tree with 5 levels? (1 point)

3- Given the following binary trees:



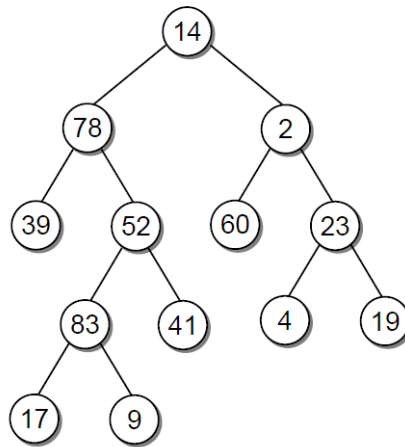
(a) Indicate all the structure properties that apply to each tree: *full*, *perfect*, *complete* (1 points).

(b) Determine the size of each tree (1 points).

(c) Determine the height of each tree (1 points).

(d) Determine the width of each tree (1 points).

4- Consider the following binary tree:



(a) Show the order the nodes will be visited in a:

i. preorder traversal (1 point)

ii. inorder traversal (1 point)

iii. postorder traversal (1 point)

iv. breadth-first traversal (1 point)

(b) Identify all the leaf nodes (1 point).

(c) Identify all the interior nodes (1 point).

(d) List all the nodes on level 4 (1 point).

(e) List all the nodes in the path to each of the following nodes (1 points):

i. 83

ii. 39

(f) Consider node 52 and list the node's (1 points):

i. descendants

ii. ancestors

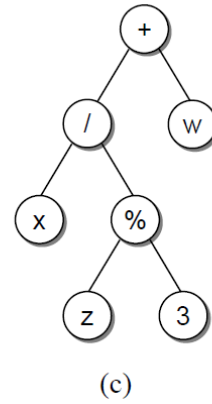
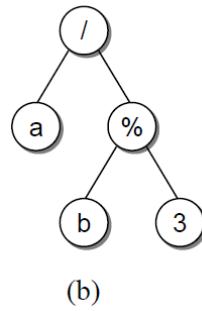
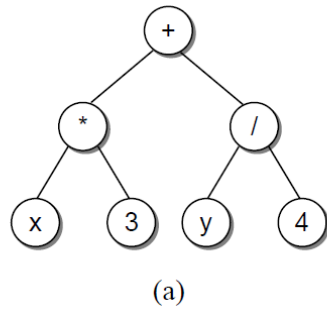
iii. Siblings

(g) Identify the depth of each of the following nodes (1 points):

i. 78

ii. 41

- 5- Determine the arithmetic expression represented by each of the following expression trees (3 points):



6- Build the expression tree for each of the following arithmetic expressions (3 points):

(a) $(A * B) / C$

(b) $A - (B * C) + D / E$

(c) $V * W \% X + Y - Z$

7- Consider the following set of values and use them to build a heap by adding one value at a time in the order listed:

30 63 2 89 16 24 19 52 27 9 4 45

(a) min-heap (1 point)

(b) max-heap (1 point)

8- Prove or show that the worst-case time of the extraction operation on a heap implemented as an array is $O(\log n)$. (1 point)

9- Prove or show that the insertion and extraction operations on a heap implemented as a Python list is $O(n)$ in the worst case. Also show that each operation has an amortized cost of $O(\log n)$. (1 point)